

ZS ASSOCIATES: SALES FORCE SIZING

In the summer of 2006, Nick Thomas faced the task of impressing his superiors at ZS Associates and securing a full-time job offer with the firm before he finished his summer internship. He had taken the lead on many extra initiatives, such as brown-bag training (a form of employee-initiated training) and organizing the intracompany tennis tournament. But for obvious reasons, he was most concerned about completing his project with high quality in the required time frame.

Thomas was working on a sales force sizing project and knew the different methodologies that could be used to arrive at the proper sales force number. His client, Santoze Pharmaceuticals Inc., was about to launch a drug named Obetor that had the potential to be a blockbuster, and had approached ZS to help them determine the most appropriate sales force size. Thomas's first internal meeting with the ZS principal on the project was in a week, and he thought it might be useful to prepare a rough estimate of the optimal sales force size using available historical data.

Company Background

ZS Associates, Inc., was founded in 1983 by Andris A. Zoltners and Prabhakant Sinha, professors at the Northwestern University's Kellogg School of Management. By 2006, the company had established itself as a global management consulting firm specializing in sales and marketing strategy, operations, and execution. ZS was recognized for its expertise in go-to-market and sales force strategies and its downstream implementation capabilities. In 2006, ZS had 850 employees working across 14 offices in the United States, Canada, Europe, Japan, and India. ZS focused on helping clients address issues related to sales and marketing such as go-to-market strategy, value proposition development, market segmentation and targeting, sales and marketing effectiveness, marketing research, sales forecasting, sales force design, marketing mix optimization, compensation design, and call planning along with sales administration.

This case was written by Sameer Kumar (MBA '07) and Arya Kalla (MBA '07) under the supervision of Robert E. Spekman, Tayloe Murphy Professor of Business Administration, with the help of ZS Associates, Inc., a global consulting firm specializing in sales and marketing. It was written as a basis for class discussion rather than to illustrate effective or ineffective handling of an administrative situation. A supplementary spreadsheet (UVA-M-0745X) is provided with this case. Copyright © 2006 by the University of Virginia Darden School Foundation, Charlottesville, VA. All rights reserved. To order copies, send an e-mail to sales@dardenbusinesspublishing.com. No part of this note may be reproduced, stored in a retrieval system, used in a spreadsheet, or transmitted in any form or by any means—electronic, mechanical, photocopying, recording, or otherwise—without the written permission of the Darden School Foundation.

ZS clientele included large multinational corporations throughout the world, as well as entrepreneurial companies with significant investment in sales and marketing. Primary clients included Global 500 companies in more than 25 industries, whom the company served at the global, regional, and local levels. Key industry sectors included: pharmaceuticals and biotech, transportation, consumer products, financial services, manufacturing, media and information, medical products/services, and information technology. Due to ZS's analytically rigorous and data intensive method of consulting, the pharmaceutical industry became a natural fit for the company and by 2006 a large portion of its clients were from this industry.

The Pharmaceutical Industry

Global pharmaceutical sales grew at a CAGR of 10.6% from 1998–2005 and worldwide audited pharmaceutical sales reached \$566 billion in 2005. The 2005 sales figures for the top 10 pharmaceutical companies and the relative change in their sales from 2004 are shown in **Table 1**.

Table 1. Sales and sales growth of top global pharmaceutical companies.

Company	Global 2005 Sales (US\$B)	% Growth from 2004
Pfizer	44.3	–4%
GlaxoSmithKline	33.9	8%
Sanofi-aventis	32.3	–5%
Novartis	24.9	16%
AstraZeneca	23.9	12%
Johnson & Johnson	22.3	1%
Merck	22.0	2%
Wyeth	15.3	10%
Bristol-Myers Squibb	15.2	–1%
Eli Lilly	14.7	12%
Total	248.8	6%

Source: “7th Annual Report on the World’s Top 50 Pharmaceutical Companies,” *Pharmaceutical Executive*, May 2005.

The pharmaceutical industry is unique in many ways. It is an industry with very high fixed costs and relative low variable costs. R&D expenses contribute a significant portion of the fixed cost. It could take up to 12 years with investments exceeding \$900 million to develop and launch a drug. To ensure that companies have the motivation to make investments for new drugs, the U.S. Food and Drug Administration (FDA) provides patent protection to companies helping them charge prices commensurate to their investments.

The Race to Profits

Most companies are very top line-driven because of high margins (often greater than 85%) and limited exclusivity for their products. It is important to note that patents can provide protection for up to 20 years. Orphan drugs, drugs that treat rare diseases affecting fewer than 200,000 Americans, are granted even longer protection from competitive threats. To encourage the development of such drugs, companies are given seven years of marketing exclusivity and are granted tax advantages for the clinical research they have done. Put simply, patents can be thought of as a means to encourage a race to get to market fastest with new drugs. Blockbuster products often drive the marketing plan because these products, with sales into billions of dollars, define success for the firm. Indeed, one study showed that for each day a drug was delayed in its commercialization, millions of dollars in lost revenue would not be uncommon.

The Powerful Few

Adding to the complexity of the industry is the fact that the buying process is very complex and contains a number of influencers, gatekeepers and others who can affect the decision-making process. First, there is the physician who plays a key role in the decision-making process because he or she prescribes the drug and is an important point of contact for the drug salesperson. Other members that have a say are the pharmacists, managed health care providers, benefit management companies, the government, and the patient who consumes the drug. All play a role in the decision-making process and some have veto power making it very easy or difficult for patients in a managed care environment to have access to a drug.

As in a number of industries, a small number of decision-makers drive the majority of revenues (the 80–20 rule). Among the 600,000 or so physicians in the United States, approximately 100,000 physicians write the majority of prescriptions across all drug categories. This is a very small number relative to the millions of people taking prescription drugs. Specialty markets composed of people with rare diseases are even more concentrated. Coupled with high margins, this phenomenon makes each patient worth thousands of dollars in terms of profits. Because the physician is a key gatekeeper, there is a strong desire to have personal contact with him or her. Even with the rise of managed care and formularies that dictate what drugs patients in a managed care setting has access to, the most expensive and often most effective tool for marketing is still the sales rep “detailing” products to physicians. This is true even though the physician does not benefit financially from the sales of the drug.

Client Background

In 2006, Santoz Pharmaceuticals employed more than 20,000 people and had annual revenues of approximately \$3 billion. Santoz had products in seven therapeutic areas: cardiovascular, thrombosis, metabolic disorders, oncology, central nervous system, vaccines, and

internal medicine. As of August 2005, Santoze had a total of 128 compounds in development in those seven therapeutic areas.

Santoze was preparing to launch a new drug (Obetor) as a treatment for obesity as well as tobacco addiction. There were high expectations for this product given its medical profile and superiority versus existing competitors. A two-year clinical trial also revealed reduced triglyceride levels, an increase of good cholesterol (HDL) and improved insulin sensitivity. The drug had been referred to as “the magic pill.” Obetor was believed to be significantly better than the competing drugs already in the market. Yet, the launch of the product could be tainted if the sales force used to take the drug to market was not sized correctly.

Project Problem

For the purpose of the first internal meeting, Thomas had to prepare a ballpark estimate of the sales force size that the client would need to promote its new drug to the specialty physician audience. He did not have any historical data on sales for this particular drug since it was a new drug. He did, however, have data on approximately 5,000 physicians that the client would want to target for the new drug. (The supplementary spreadsheet that accompanies this case, UVA-M-0745X, contains data on the total number of prescriptions that each of the target physicians had written in the previous year for the obesity market.) This data would help him group the physicians into deciles and thus would give him the number of physicians in each decile. Deciles are formed by first arranging the physicians in descending order in terms of the revenue (prescriptions) he or she generated and then grouping the physicians into 10 equal revenue (prescription) generating groups (i.e., the cumulative prescriptions of the physicians in any decile will be equal to the cumulative prescriptions of the physicians in any of the other deciles). Typically, the first decile has the least number of doctors and the 10th decile has the largest number of doctors.

He also needed to determine the number of times a physician should be called on and the expected value of contribution he could expect per physician. These numbers would differ for physicians depending on the decile segment. Using historical data for this particular drug segment, Thomas approximated these numbers, which are found in **Exhibit 1**.

Thomas knew that an average salesperson could make eight sales calls in one day and worked roughly 200 days per year. Thomas also knew that, on average, a salesperson’s fully loaded costs were \$175,000 per year. To hire and train a salesperson tended to be expensive and Thomas estimated that training and recruiting cost often equaled \$50,000 in the first year. From all this information, Thomas thought he should be able to determine the size of the sales force required to meet the company’s 150% return on investment threshold. He knew that he had to factor in carryover, and approximated 70% of first-year sales to carry over to second year and 70% of second-year sales to carry over to third year. Future sales would also need to be discounted to get their present value. Thomas thought it would be appropriate to use the industry’s average cost of capital (12%) for discounting sales. He also knew that he should

conduct multiple analysis, each analysis assuming a different time-horizon of maintaining the sales force, because the training and recruiting costs would be a one-time investment rather than an ongoing expense; however, given his time constraints for the meeting, he thought he would calculate assuming that the sales force would be hired for only one year.

Exhibit 1

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Expected Calls Required and Value of Contribution per Physician in Different Deciles

Deciles	Calls required per physician	Expected value of contribution per physician*
1	40	\$13,000
2	38	\$12,000
3	34	\$10,000
4	32	\$8,500
5	30	\$7,200
6	26	\$6,200
7	20	\$5,000
8	18	\$3,500
9	16	\$2,500
10	12	\$2,000

*Expected value is the gross profit, not taking into account the sales force costs.